
4.4 End of Chapter Problems

Problem 1

Choose a real number uniformly at random in the interval $[2, 6]$ and call it X .

- Find the CDF of X , $F_X(x)$.
 - Find EX .
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Problem 2

Let X be a continuous random variable with the following PDF

$$f_X(x) = \begin{cases} ce^{-4x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

where c is a positive constant.

- Find c .
 - Find the CDF of X , $F_X(x)$.
 - Find $P(2 < X < 5)$.
 - Find EX .
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Problem 3

Let X be a continuous random variable with PDF

$$f_X(x) = \begin{cases} x^2 + \frac{2}{3} & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

- Find $E(X^n)$, for $n = 1, 2, 3, \dots$.
 - Find the variance of X .
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Problem 4

Let X be a *uniform*(0, 1) random variable, and let $Y = e^{-X}$.

- Find the CDF of Y .
 - Find the PDF of Y .
 - Find EY .
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Problem 5

Let X be a continuous random variable with PDF

$$f_X(x) = \begin{cases} \frac{5}{32}x^4 & 0 < x \leq 2 \\ 0 & \text{otherwise} \end{cases}$$

and let $Y = X^2$.

- Find the CDF of Y .
 - Find the PDF of Y .
 - Find EY .
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Problem 6

Let $X \sim \text{Exponential}(\lambda)$, and $Y = aX$, where a is a positive real number. Show that

$$Y \sim \text{Exponential}\left(\frac{\lambda}{a}\right).$$

Problem 7

Let $X \sim \text{Exponential}(\lambda)$. Show that

- a. $EX^n = \frac{n}{\lambda} EX^{n-1}$, for $n = 1, 2, 3, \dots$;
 - b. $EX^n = \frac{n!}{\lambda^n}$, for $n = 1, 2, 3, \dots$.
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Problem 8

Let $X \sim N(3, 9)$.

- a. Find $P(X > 0)$.
 - b. Find $P(-3 < X < 8)$.
 - c. Find $P(X > 5 | X > 3)$.
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Problem 9

Let $X \sim N(3, 9)$ and $Y = 5 - X$.

- a. Find $P(X > 2)$.
 - b. Find $P(-1 < Y < 3)$.
 - c. Find $P(X > 4 | Y < 2)$.
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Problem 10

Let X be a continuous random variable with PDF

$$f_X(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{x^2}{2}} \quad \text{for all } x \in \mathbb{R}.$$

and let $Y = \sqrt{|X|}$. Find $f_Y(y)$.

Problem 11

Let $X \sim \text{Exponential}(2)$ and $Y = 2 + 3X$.

- a. Find $P(X > 2)$.
 - b. Find EY and $\text{Var}(Y)$.
 - c. Find $P(X > 2 | Y < 11)$.
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Problem 12

The median of a continuous random variable X can be defined as the unique real number m that satisfies

$$P(X \geq m) = P(X < m) = \frac{1}{2}.$$

Find the median of the following random variables

- a. $X \sim \text{Uniform}(a, b)$.
 - b. $Y \sim \text{Exponential}(\lambda)$.
 - c. $W \sim N(\mu, \sigma^2)$.
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Problem 13

Let X be a random variable with the following CDF

$$F_X(x) = \begin{cases} 0 & \text{for } x < 0 \\ x & \text{for } 0 \leq x < \frac{1}{4} \\ x + \frac{1}{2} & \text{for } \frac{1}{4} \leq x < \frac{1}{2} \\ 1 & \text{for } x \geq \frac{1}{2} \end{cases}$$

- Plot $F_X(x)$ and explain why X is a mixed random variable.
- Find $P(X \leq \frac{1}{3})$.
- Find $P(X \geq \frac{1}{4})$.
- Write CDF of X in the form of

$$F_X(x) = C(x) + D(x),$$

where $C(x)$ is a continuous function and $D(x)$ is in the form of a staircase function, i.e.,

$$D(x) = \sum_k a_k u(x - x_k).$$

- Find $c(x) = \frac{d}{dx} C(x)$.
 - Find EX using $EX = \int_{-\infty}^{\infty} xc(x)dx + \sum_k x_k a_k$
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Problem 14

Let X be a random variable with the following CDF

$$F_X(x) = \begin{cases} 0 & \text{for } x < 0 \\ x & \text{for } 0 \leq x < \frac{1}{4} \\ x + \frac{1}{2} & \text{for } \frac{1}{4} \leq x < \frac{1}{2} \\ 1 & \text{for } x \geq \frac{1}{2} \end{cases}$$

- Find the generalized PDF of X , $f_X(x)$.
 - Find EX using $f_X(x)$.
 - Find $Var(X)$ using $f_X(x)$.
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Problem 15

Let X be a mixed random variable with the following generalized PDF

$$f_X(x) = \frac{1}{3}\delta(x+2) + \frac{1}{6}\delta(x-1) + \frac{1}{2}\frac{1}{\sqrt{2\pi}}e^{-\frac{x^2}{2}}.$$

- Find $P(X = 1)$ and $P(X = -2)$.
 - Find $P(X \geq 1)$.
 - Find $P(X = 1 | X \geq 1)$.
 - Find EX and $Var(X)$.
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Problem 16

A company makes a certain device. We are interested in the lifetime of the device. It is estimated that around 2% of the devices are defective from the start so they have a lifetime of 0 years. If a device is not defective, then the lifetime of the device is exponentially distributed with a parameter $\lambda = 2$ years. Let X be the lifetime of a randomly chosen device.

- Find the generalized PDF of X .
- Find $P(X \geq 1)$.
- Find $P(X > 2 | X \geq 1)$.
- Find EX and $Var(X)$.

Problem 17

A continuous random variable is said to have a $Laplace(\mu, b)$ distribution [14] if its PDF is given by

$$f_X(x) = \frac{1}{2b} \exp\left(-\frac{|x - \mu|}{b}\right) \\ = \begin{cases} \frac{1}{2b} \exp\left(\frac{x - \mu}{b}\right) & \text{if } x < \mu \\ \frac{1}{2b} \exp\left(-\frac{x - \mu}{b}\right) & \text{if } x \geq \mu \end{cases}$$

where $\mu \in \mathbb{R}$ and $b > 0$.

- If $X \sim Laplace(0, 1)$, find EX and $Var(X)$.
 - If $X \sim Laplace(0, 1)$ and $Y = bX + \mu$, show that $Y \sim Laplace(\mu, b)$.
 - Let $Y \sim Laplace(\mu, b)$, where $\mu \in \mathbb{R}$ and $b > 0$. Find EY and $Var(Y)$.
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Problem 18

Let $X \sim Laplace(0, b)$, i.e.,

$$f_X(x) = \frac{1}{2b} \exp\left(-\frac{|x|}{b}\right),$$

where $b > 0$. Define $Y = |X|$. Show that $Y \sim Exponential\left(\frac{1}{b}\right)$.

Problem 19

A continuous random variable is said to have the standard Cauchy distribution if its PDF is given by

$$f_X(x) = \frac{1}{\pi(1 + x^2)}.$$

If X has a standard Cauchy distribution, show that EX is not well-defined. Also, show $EX^2 = \infty$.

Problem 20

A continuous random variable is said to have a Rayleigh distribution with parameter σ if its PDF is given by

$$f_X(x) = \frac{x}{\sigma^2} e^{-x^2/2\sigma^2} u(x) \\ = \begin{cases} \frac{x}{\sigma^2} e^{-x^2/2\sigma^2} & \text{if } x \geq 0 \\ 0 & \text{if } x < 0 \end{cases}$$

where $\sigma > 0$.

- If $X \sim Rayleigh(\sigma)$, find EX .
 - If $X \sim Rayleigh(\sigma)$, find the CDF of X , $F_X(x)$.
 - If $X \sim Exponential(1)$ and $Y = \sqrt{2\sigma^2 X}$, show that $Y \sim Rayleigh(\sigma)$.
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Problem 21

A continuous random variable is said to have a $Pareto(x_m, \alpha)$ distribution [15] if its PDF is given by

$$f_X(x) = \begin{cases} \alpha \frac{x_m^\alpha}{x^{\alpha+1}} & \text{for } x \geq x_m \\ 0 & \text{for } x < x_m \end{cases}$$

where $x_m, \alpha > 0$. Let $X \sim \text{Pareto}(x_m, \alpha)$.

- Find the CDF of X , $F_X(x)$.
 - Find $P(X > 3x_m | X > 2x_m)$.
 - If $\alpha > 2$, find EX and $Var(X)$.
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Problem 22

Let $Z \sim N(0, 1)$. If we define $X = e^{\sigma Z + \mu}$, then we say that X has a log-normal distribution with parameters μ and σ , and we write $X \sim \text{LogNormal}(\mu, \sigma)$.

- If $X \sim \text{LogNormal}(\mu, \sigma)$, find the CDF of X in terms of the Φ function.
 - Find EX and $Var(X)$.
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Problem 23

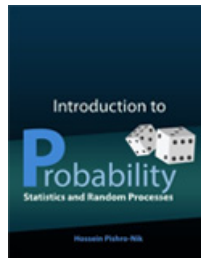
Let $X_1, X_2, \dots, X_n$ be independent random variables with $X_i \sim \text{Exponential}(\lambda)$. Define

$$Y = X_1 + X_2 + \dots + X_n.$$

As we will see later, Y has a Gamma distribution with parameters n and λ , i.e., $Y \sim \text{Gamma}(n, \lambda)$. Using this, show that if $Y \sim \text{Gamma}(n, \lambda)$, then $EY = \frac{n}{\lambda}$ and $Var(Y) = \frac{n}{\lambda^2}$.

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